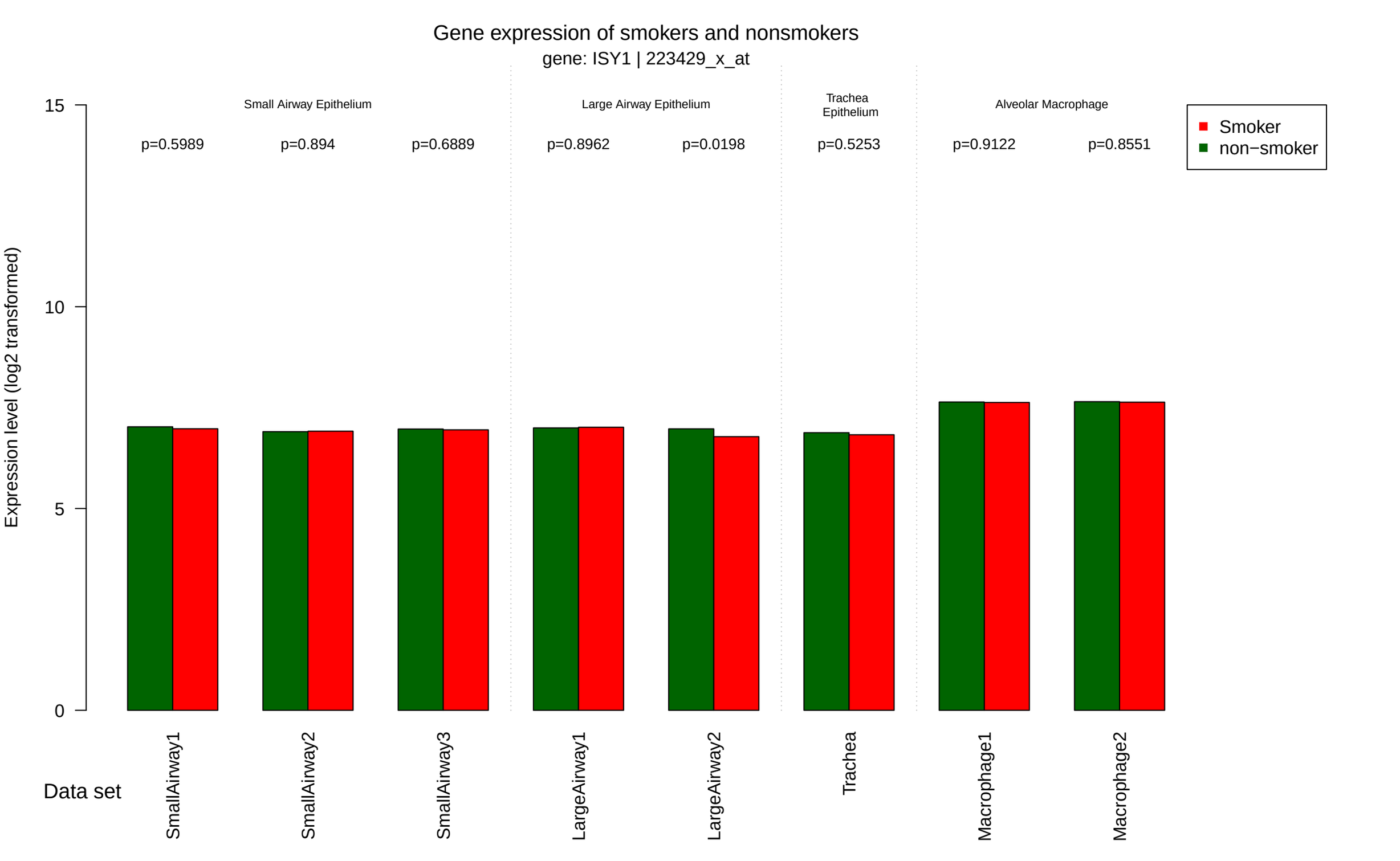
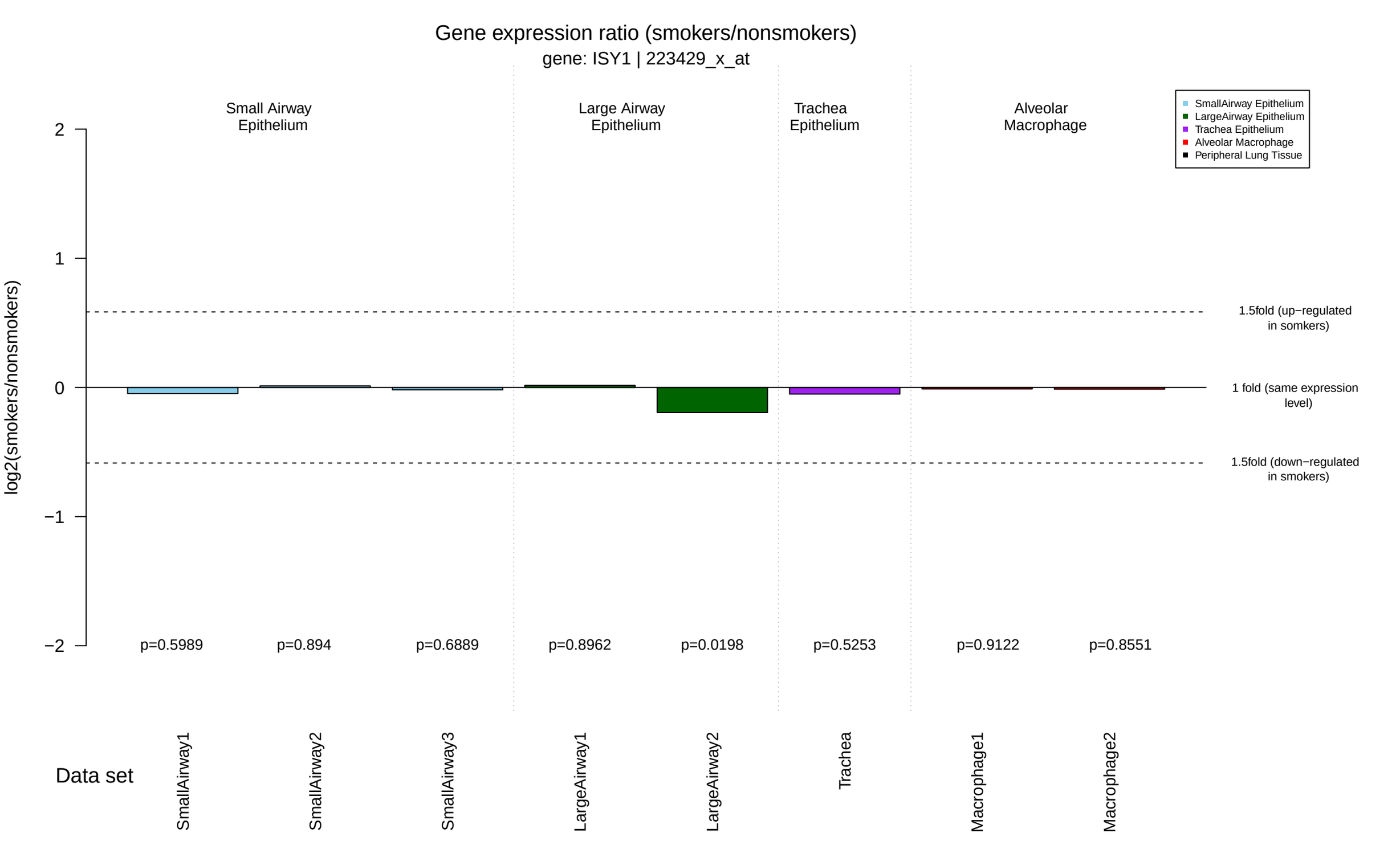
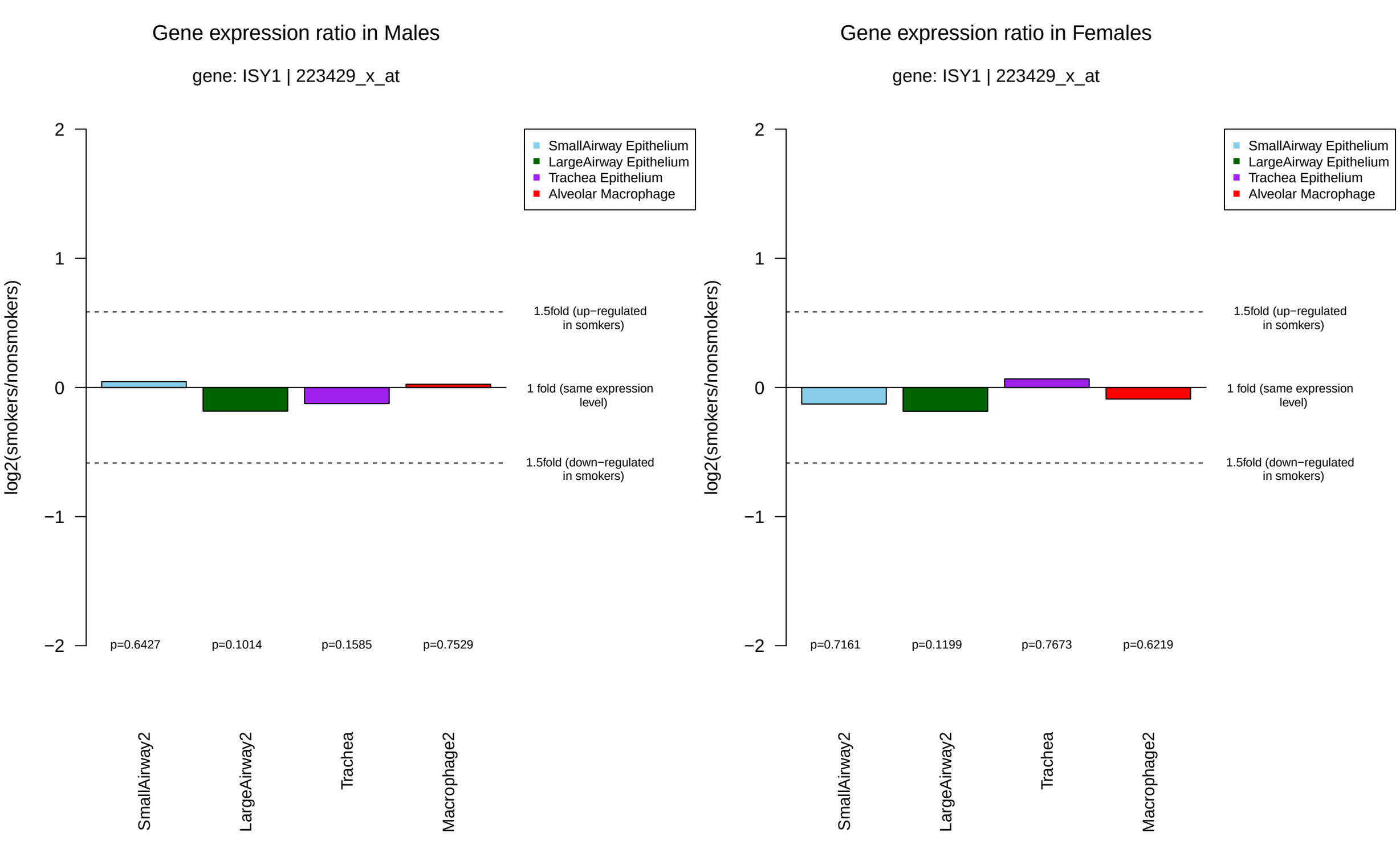
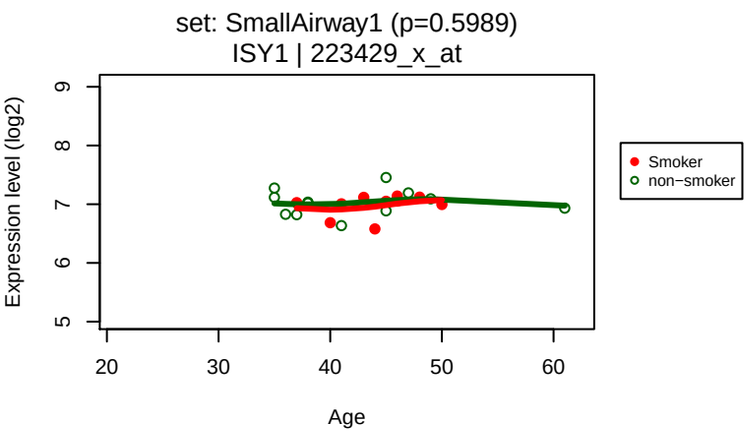
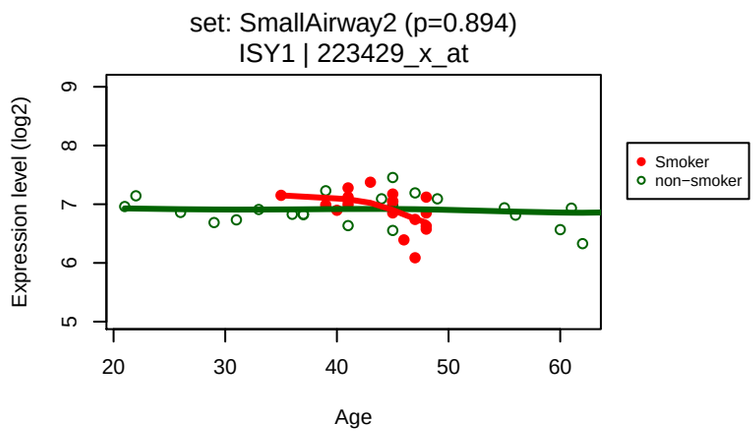
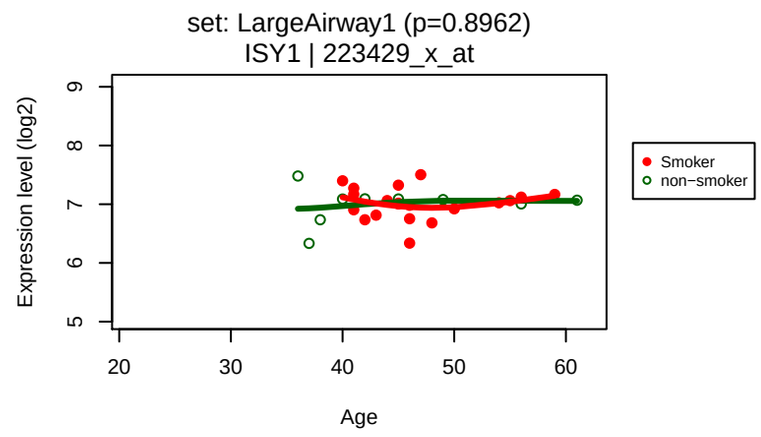

Gene expression of smokers and nonsmokers
gene: ISY1 | 223429_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5989
p=0.894
p=0.6889
p=0.8962
p=0.0198
p=0.5253
p=0.9122
p=0.8551
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

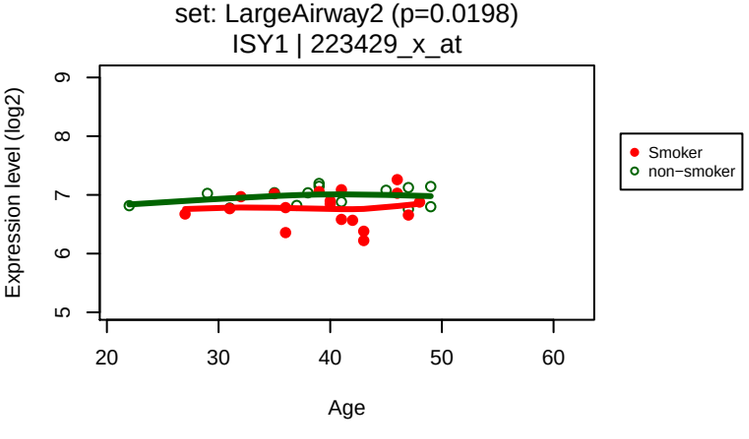

Gene expression ratio (smokers/nonsmokers)
gene: ISY1 | 223429_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5989
p=0.894
p=0.6889
p=0.8962
p=0.0198
p=0.5253
p=0.9122
p=0.8551
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

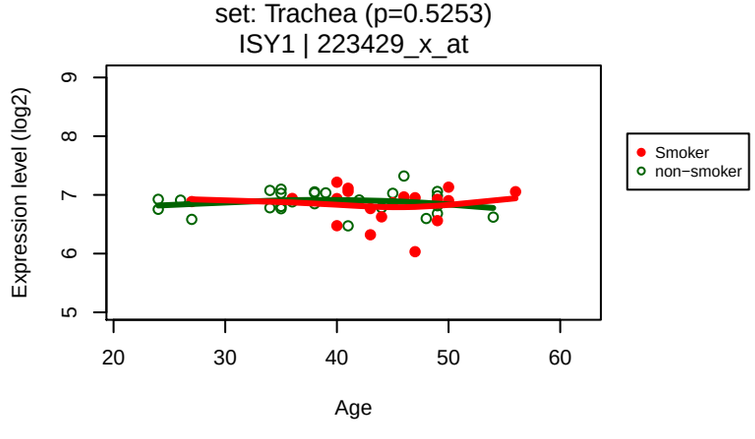

Gene expression ratio in Males
gene: ISY1 | 223429_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6427
p=0.1014
p=0.1585
p=0.7529
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISY1 | 223429_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7161
p=0.1199
p=0.7673
p=0.6219
SmallAirway2
LargeAirway2
Trachea
Macrophage2

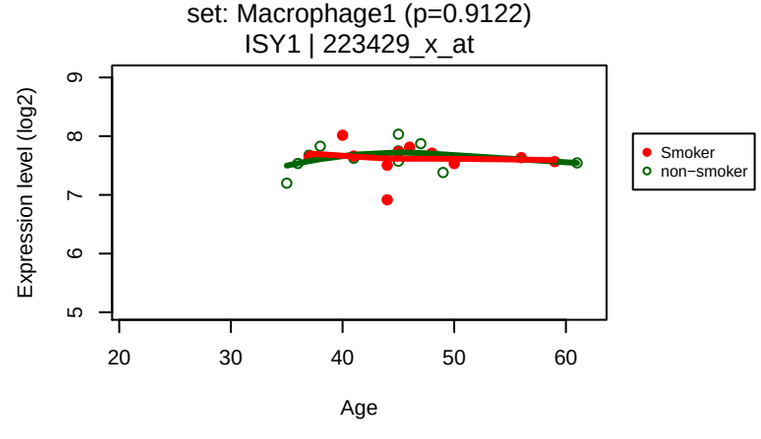

set: SmallAirway1 (p=0.5989)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker

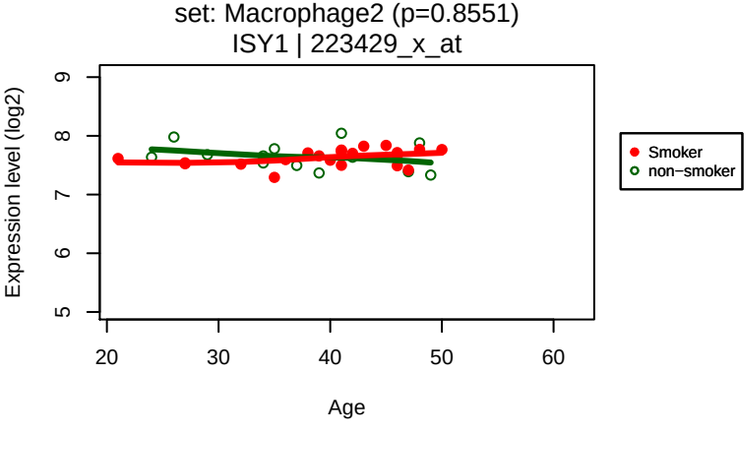

set: SmallAirway2 (p=0.894)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker

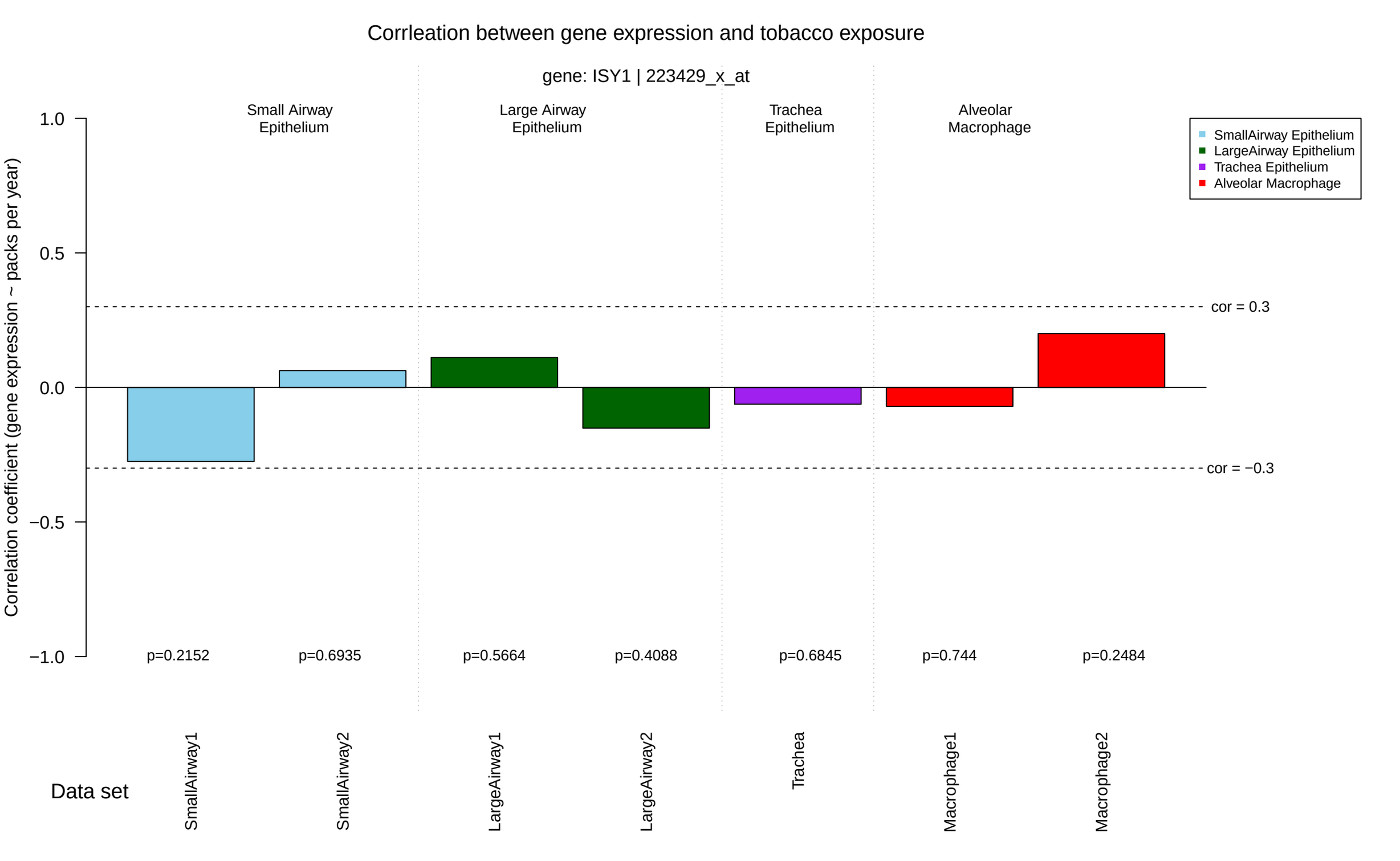

set: LargeAirway1 (p=0.8962)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0198)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5253)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9122)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8551)
ISY1 | 223429_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISY1 | 223429_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2152
p=0.6935
p=0.5664
p=0.4088
p=0.6845
p=0.744
p=0.2484
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2